Introduction to Graph Theory

Why Graphs?

Alexander Golovnev

Outline

What is a Graph?

Graph Examples

Graph Applications

Graphs

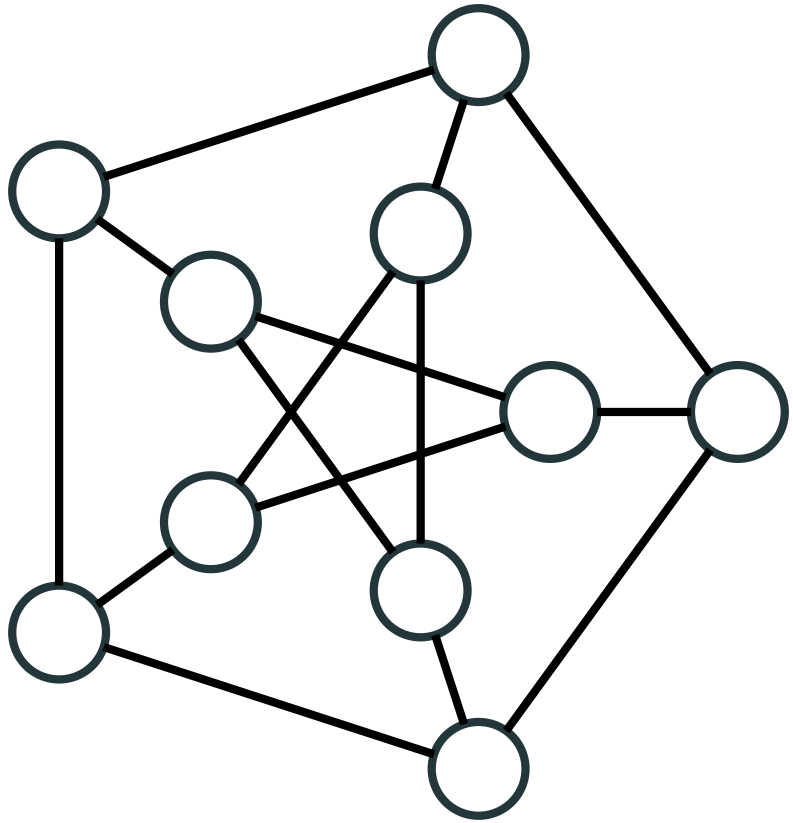
- A set of Objects

Graphs

- A set of Objects
- Relations between pairs of objects

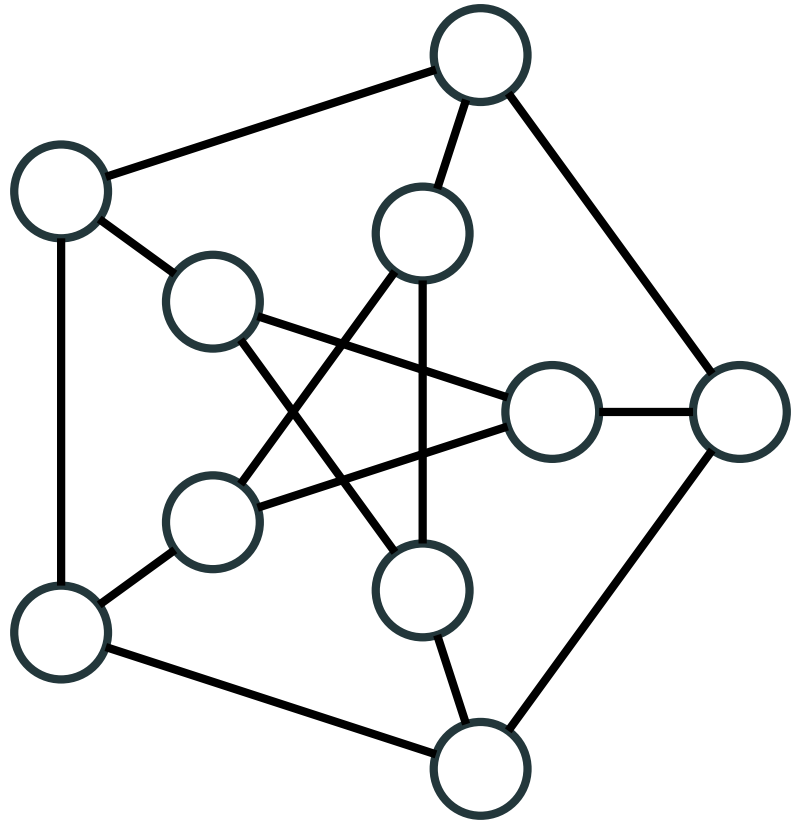
Graphs

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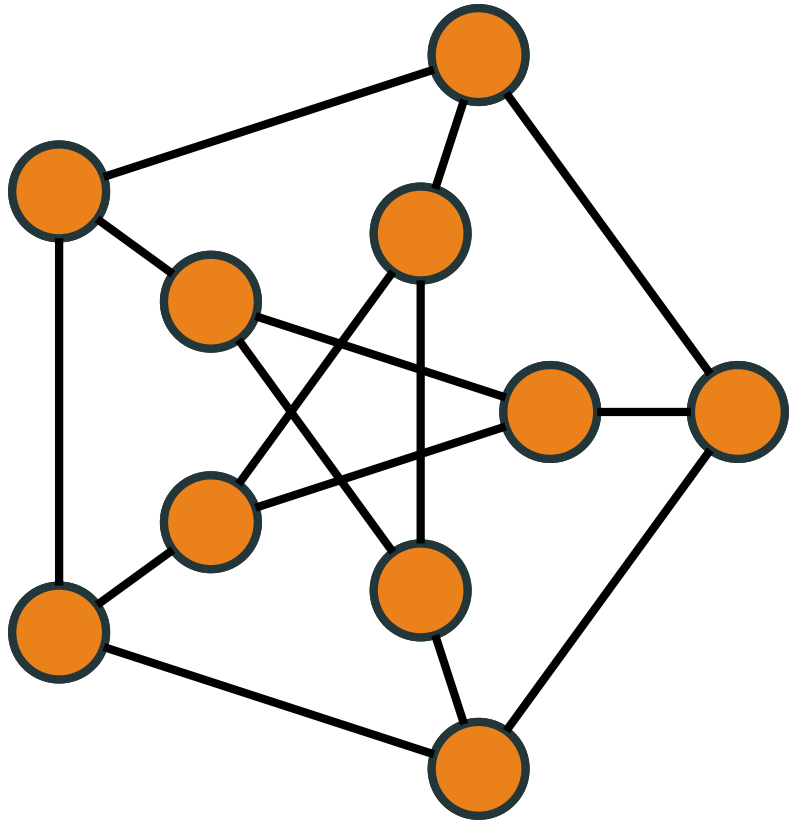
Definition

- A Graph $G = (V, E)$



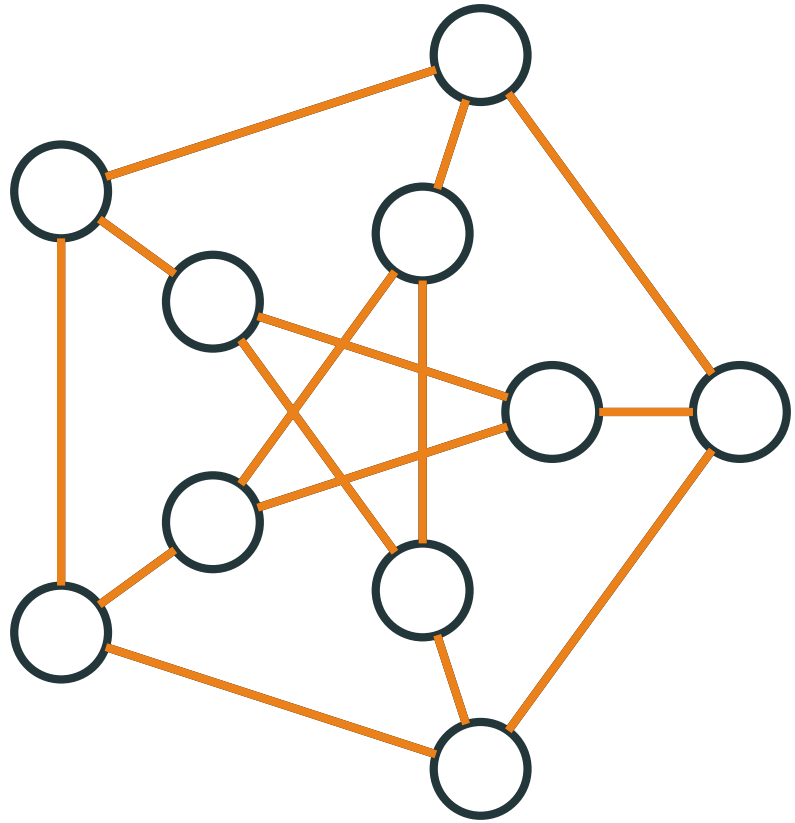
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Definition

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- A set V of Vertices/Nodes
- A set E of Edges



Vocabulary



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- *u* and *e* are Incident
- *u* and *v* are Adjacent
- *u* and *v* are Neighbors

Drawing a Graph

Objects: {A,B,C,D}

Relations: {{A,C},{D,A},{B,D},{C,B}}

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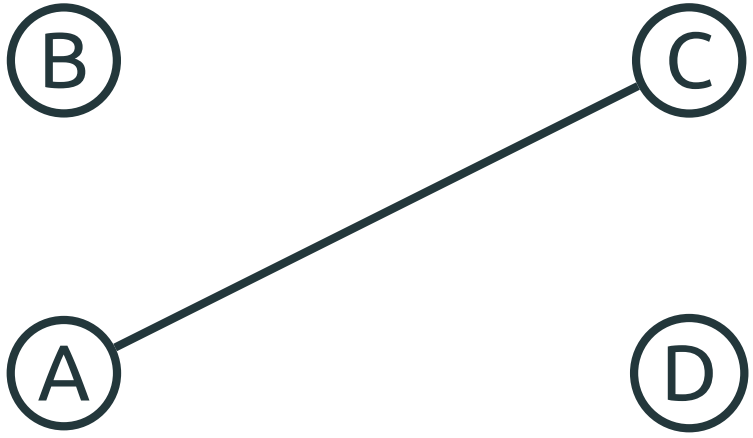
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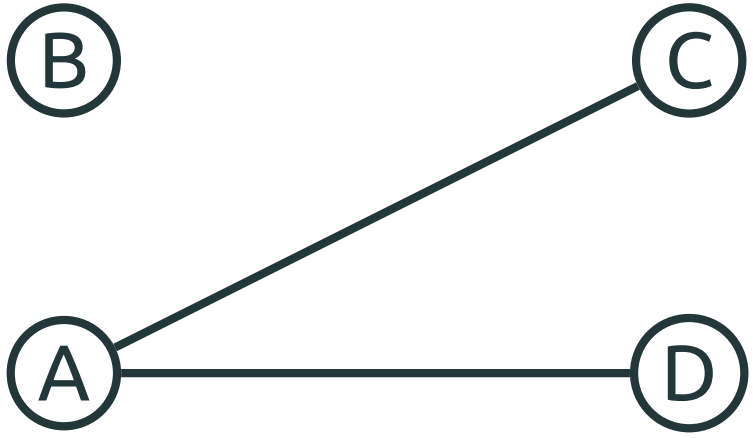
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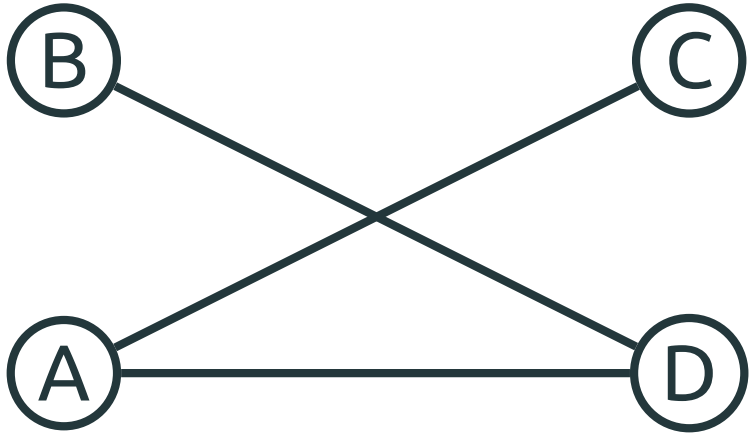
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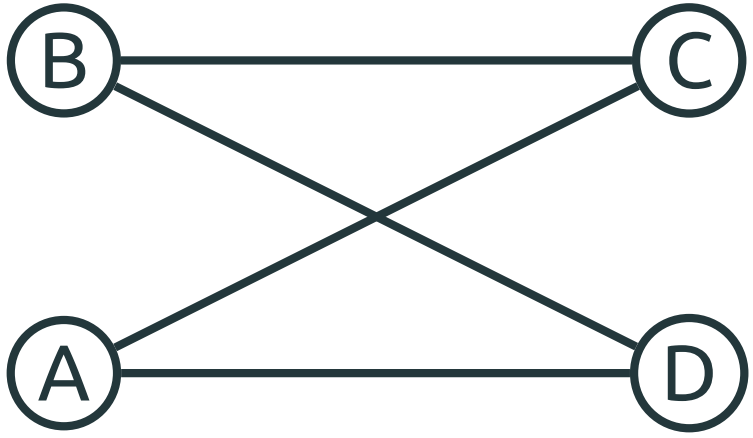
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Drawing a Graph

Objects: {A,B,C,D}

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Drawing a Graph

Objects: {A,B,C,D}

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Drawing a Graph

Objects: {A,B,C,D}

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Drawing a Graph

Objects: {A,B,C,D}

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Directed Graph



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- They describe asymmetric relations
- There is a flight from A to B, but not the other way around
- Such a graph is called Directed

Drawing a Directed Graph

Objects: {A,B,C,D}

Relations: {(A,C),(D,A),(B,D),(C,B)}



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Objects: {A,B,C,D}

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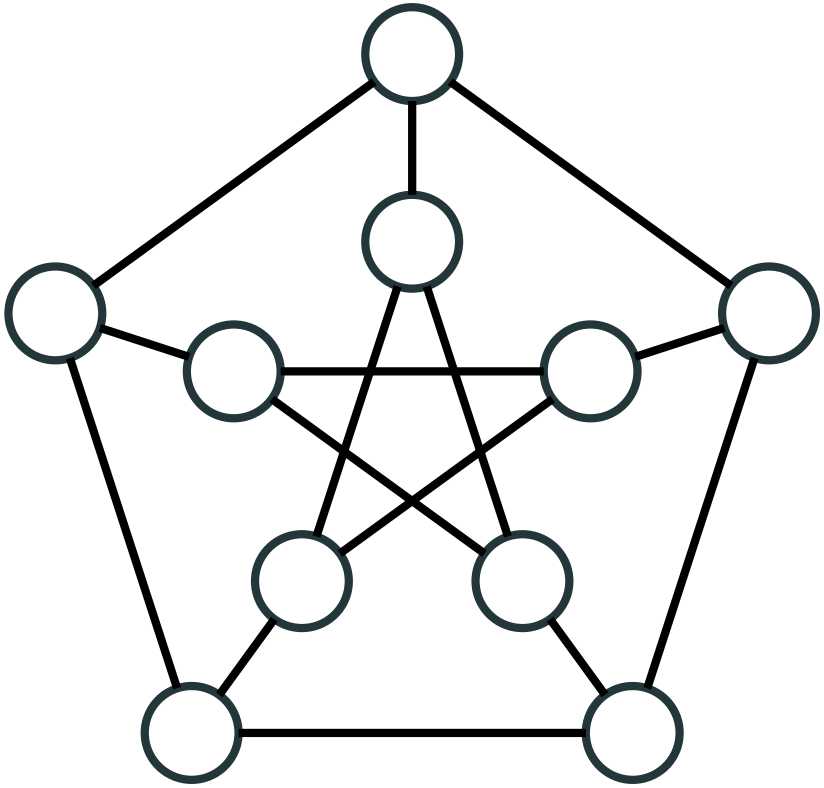
Drawing a Directed Graph

Objects: {A,B,C,D}

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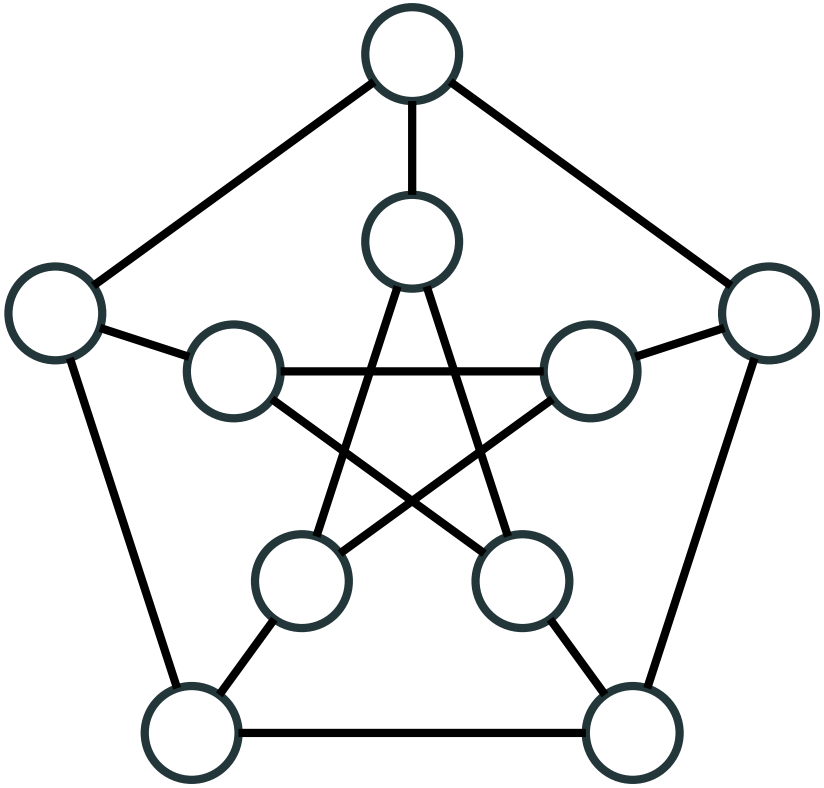


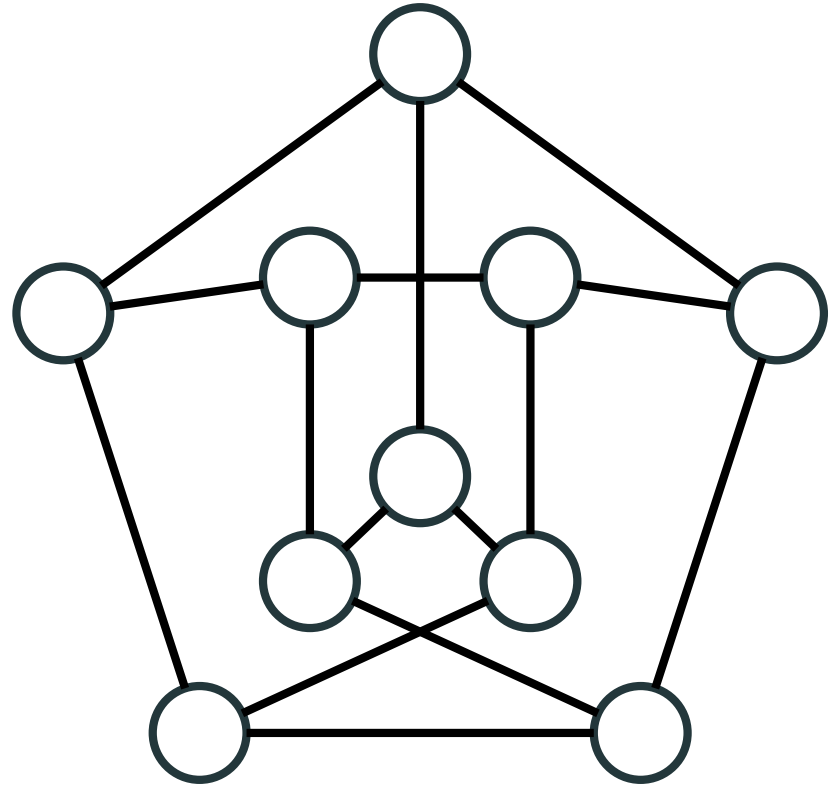
Many Ways to Draw a Graph



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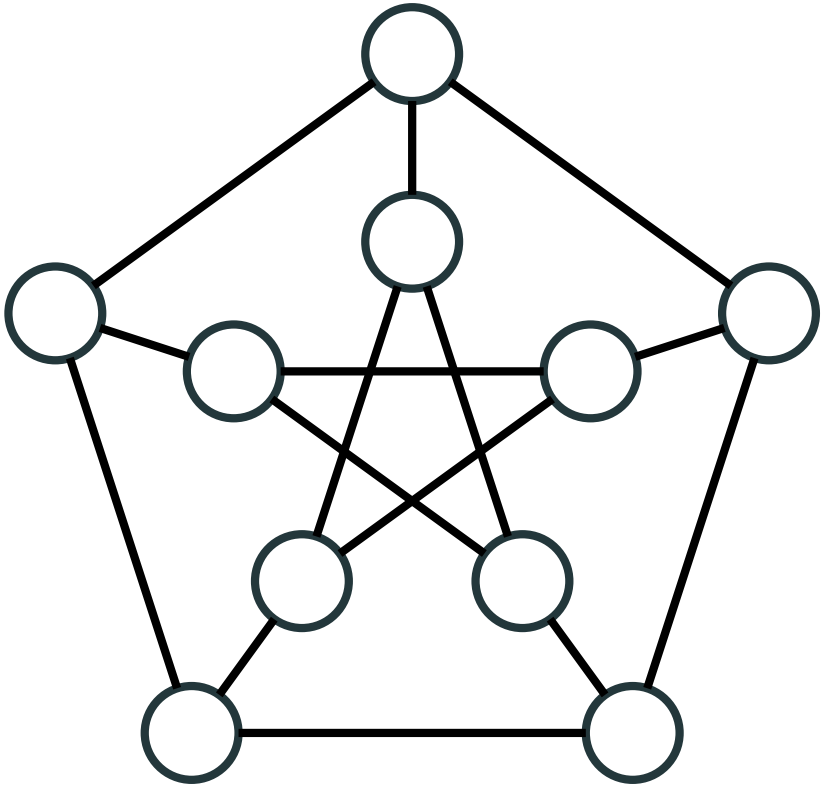
Are these graphs the same?

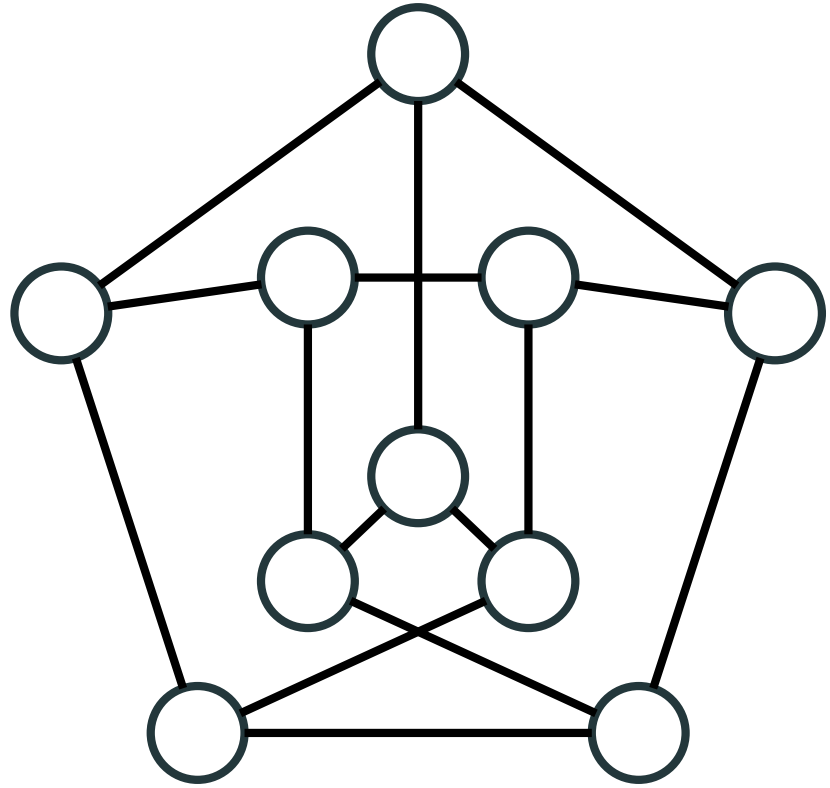




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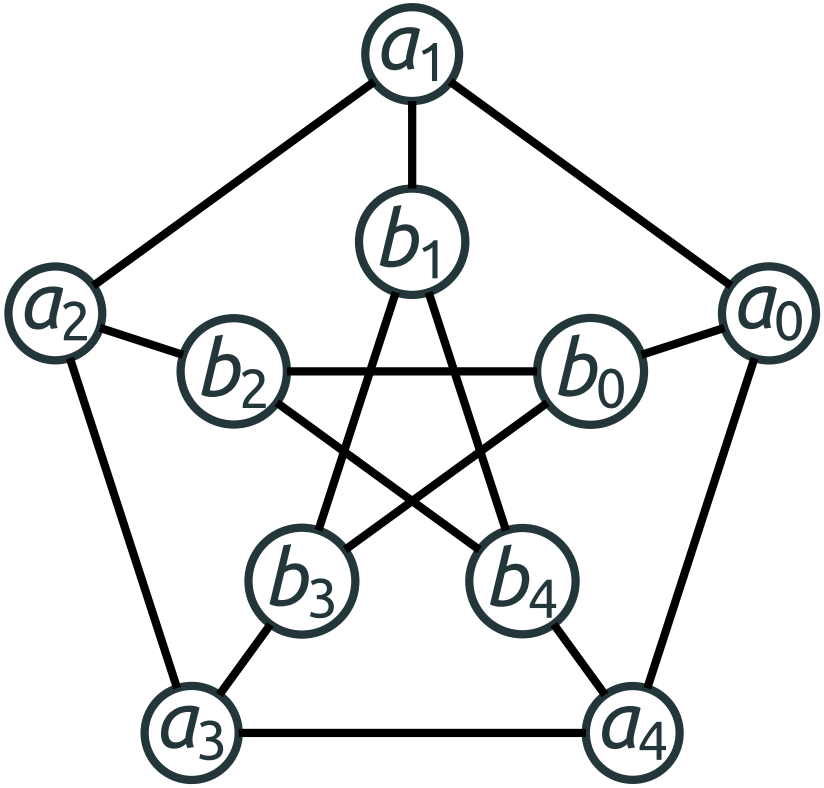
10 vertices and 15 edges?

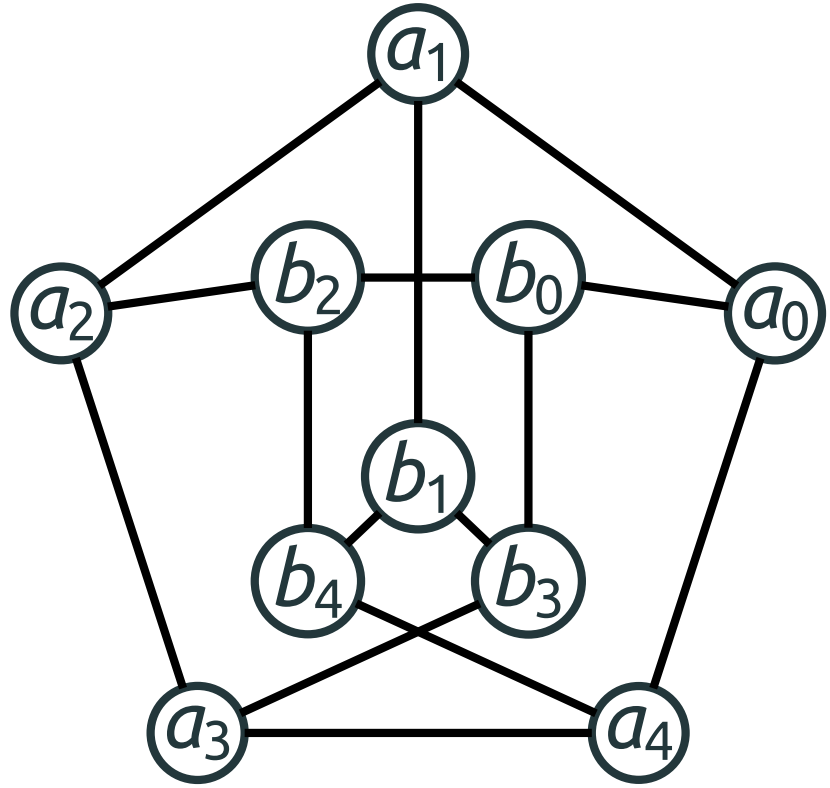




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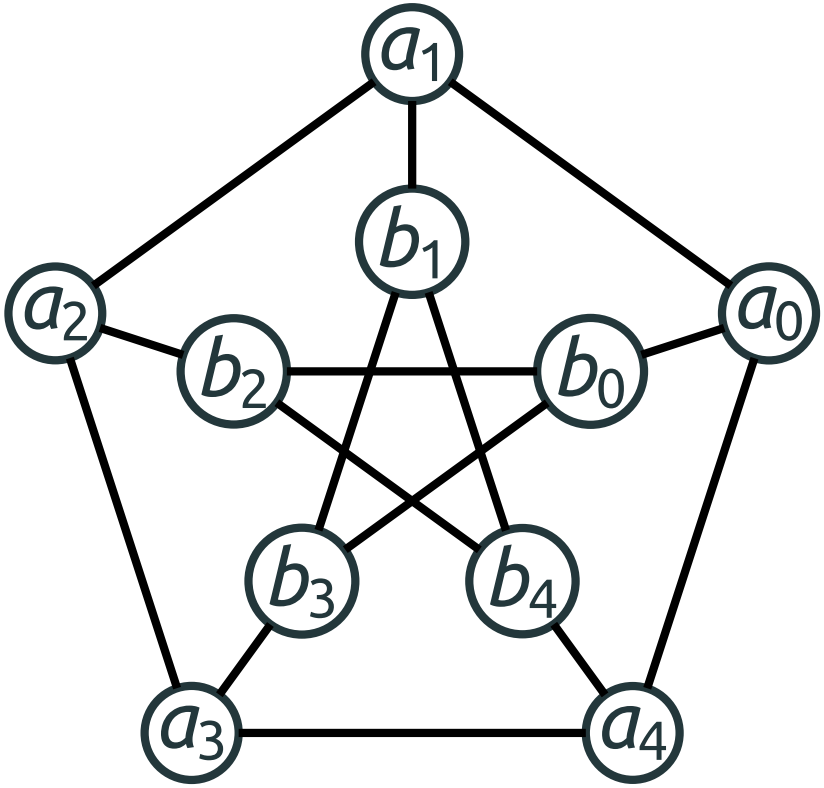
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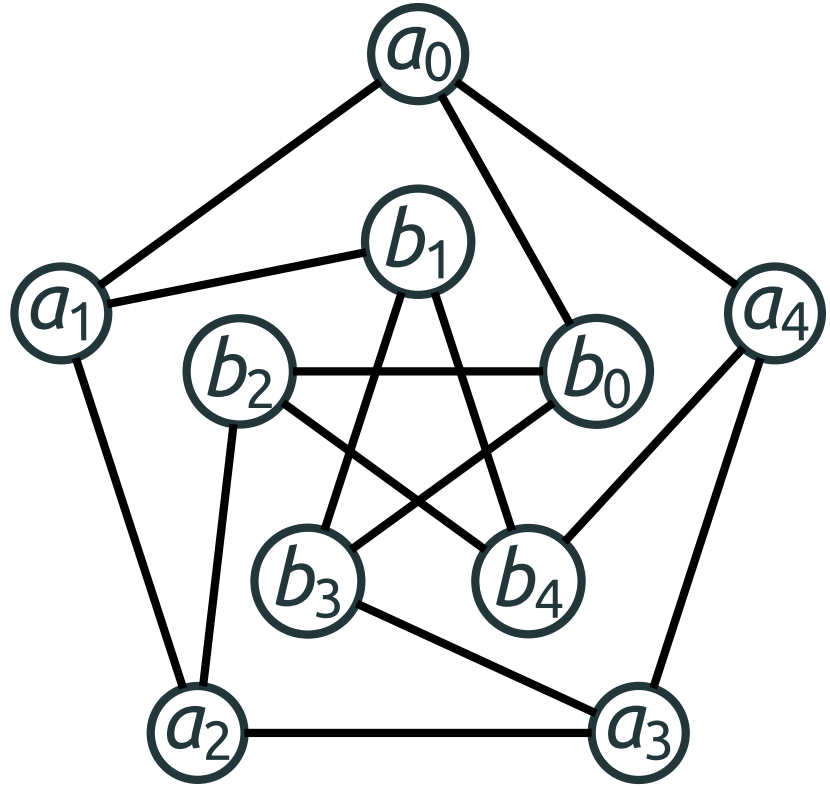




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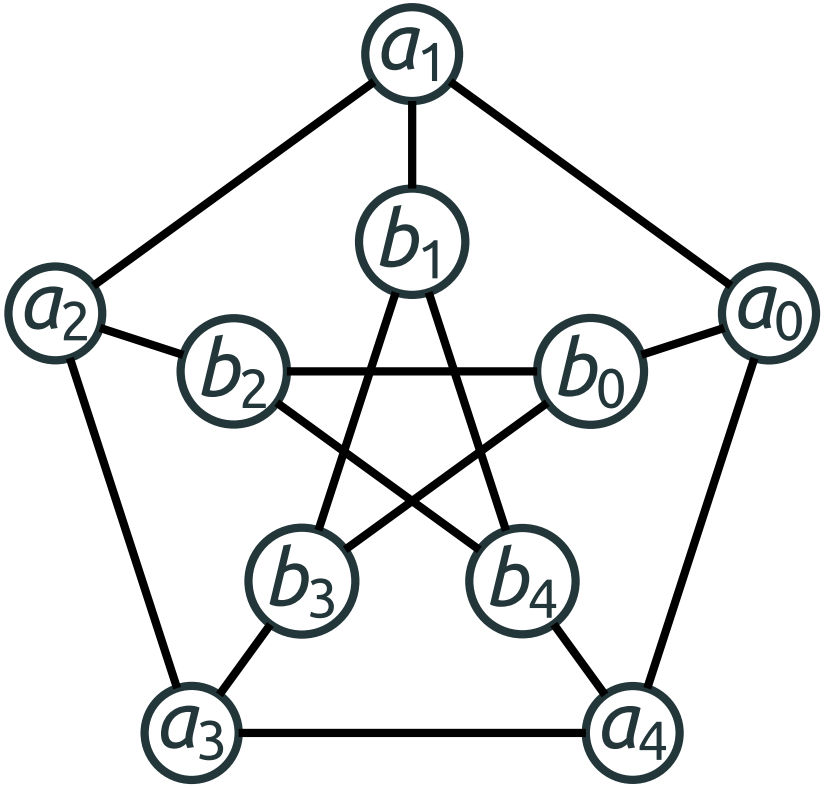
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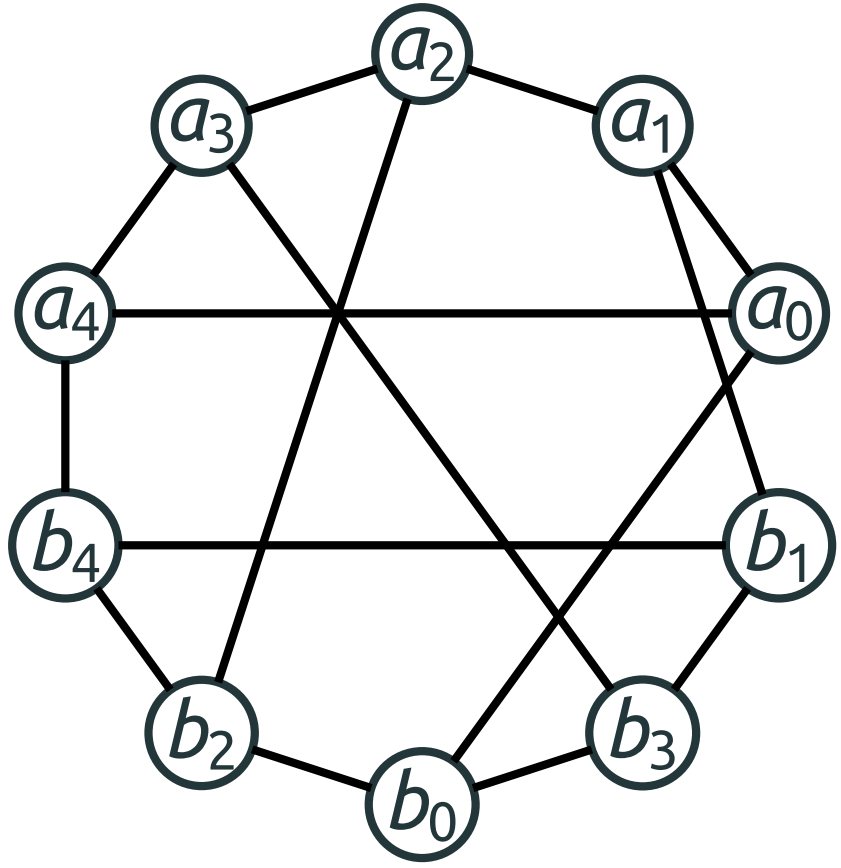




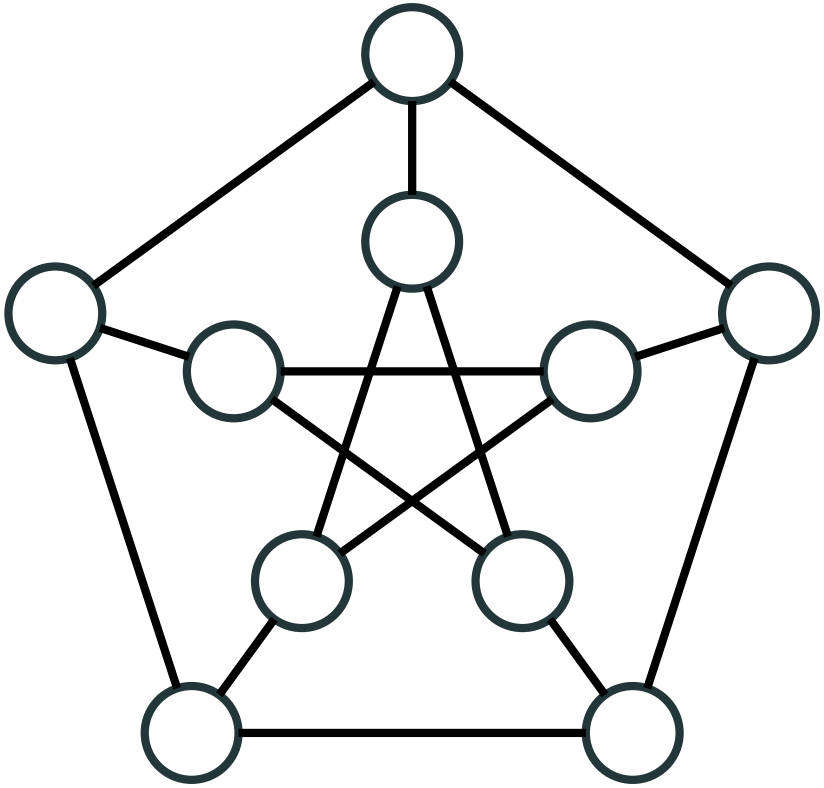
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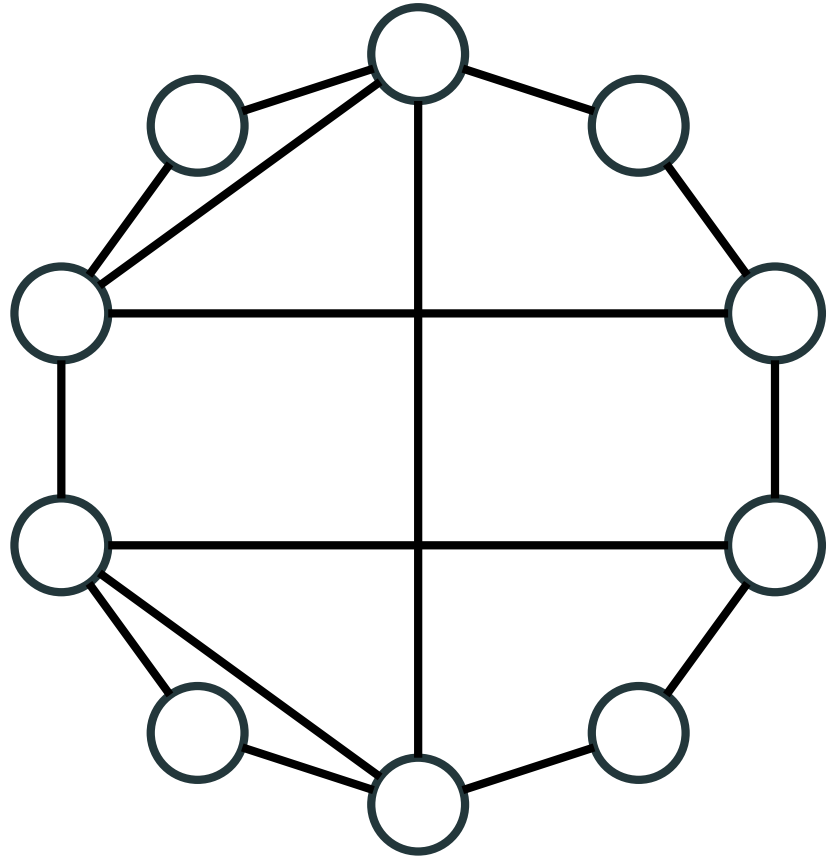
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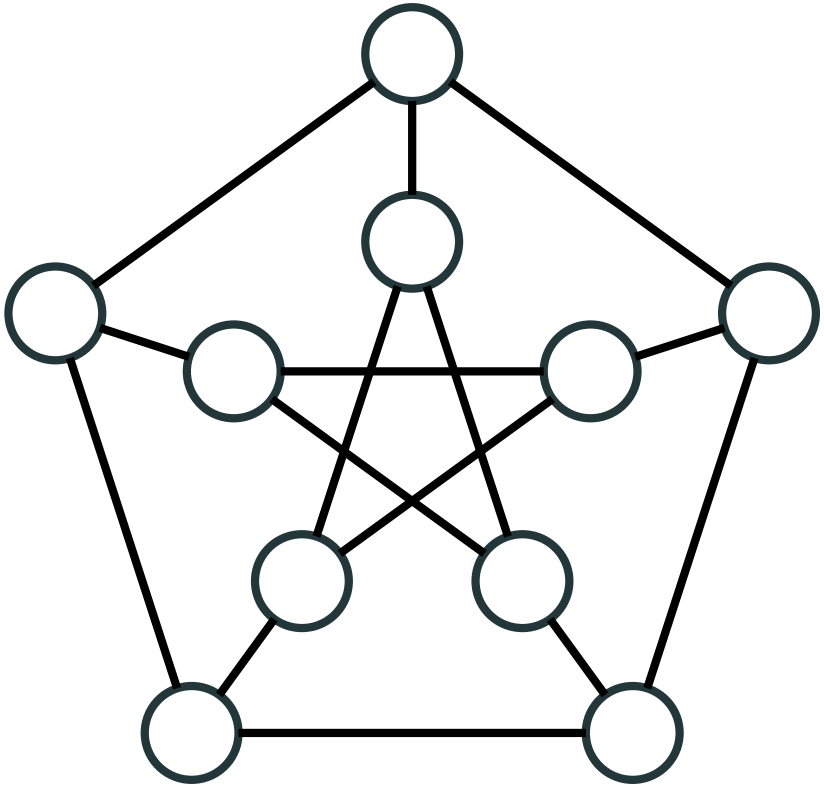
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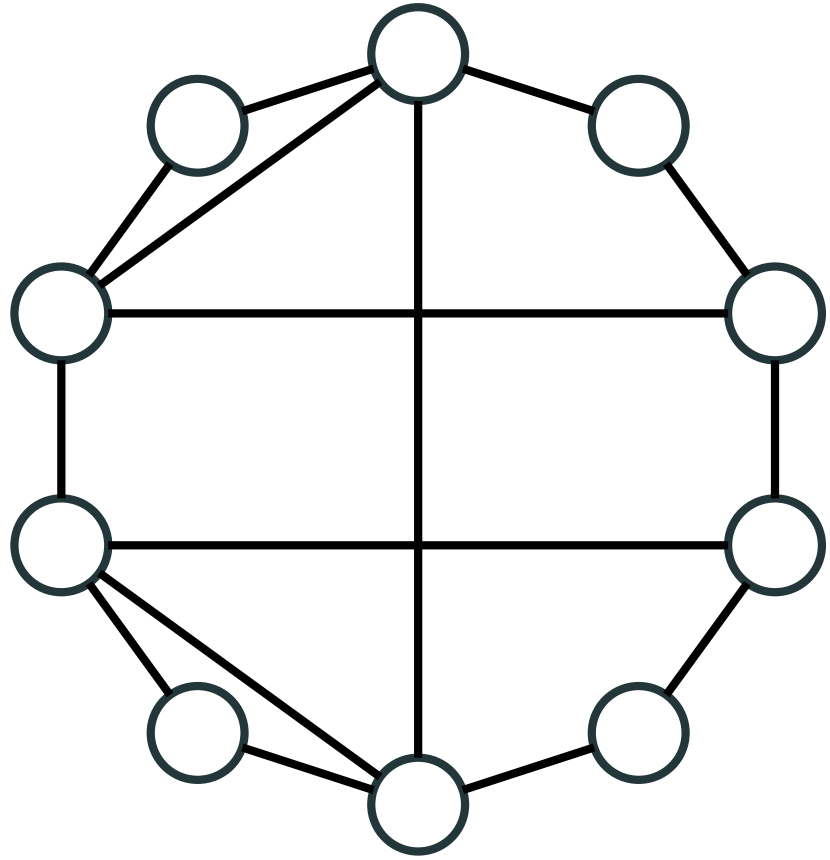




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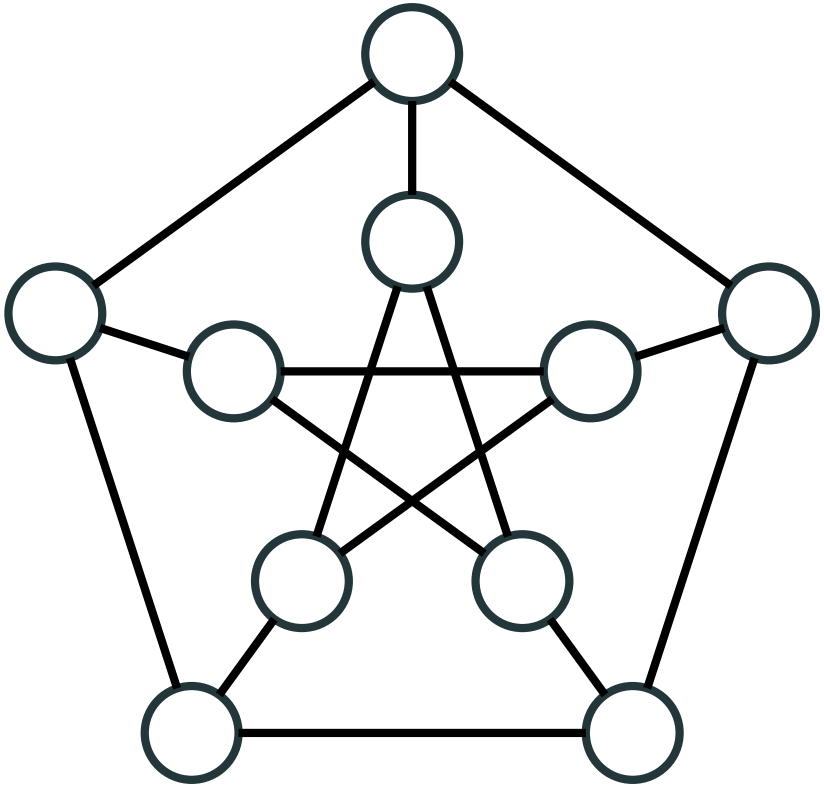
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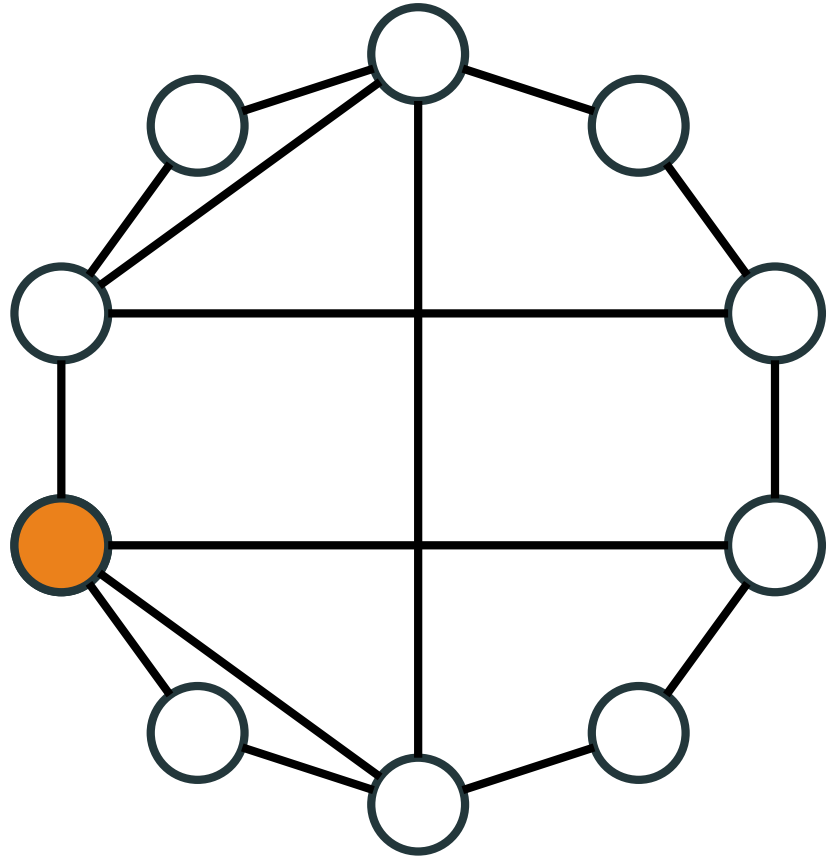




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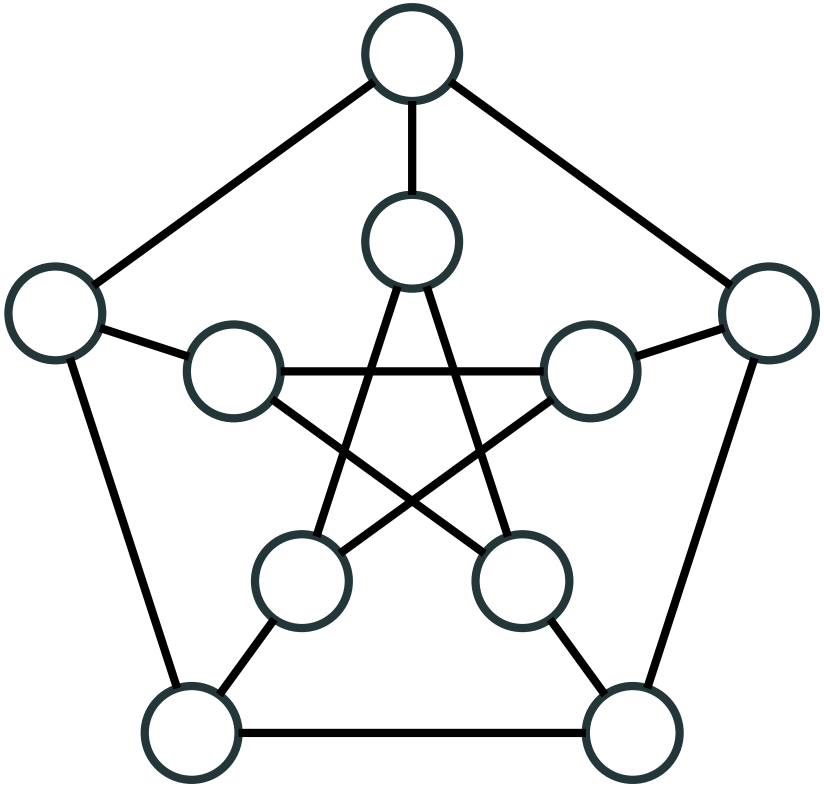
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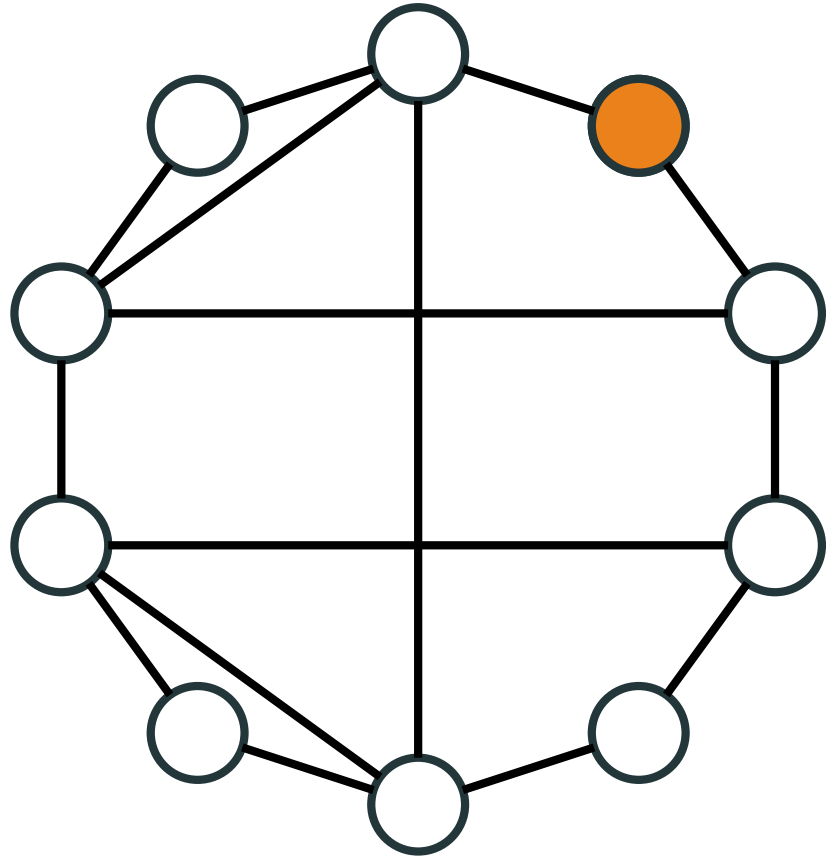




Are These Graphs the Same?

10 vertices and 15 edges?





Graph Drawing is Beautiful!



Donald E. Knuth

Graph drawing is the best possible field I can think of: It merges aesthetics, mathematical beauty and wonderful algorithms.
It therefore provides a harmonic balance between the left and right brain parts.

Outline

What is a Graph?

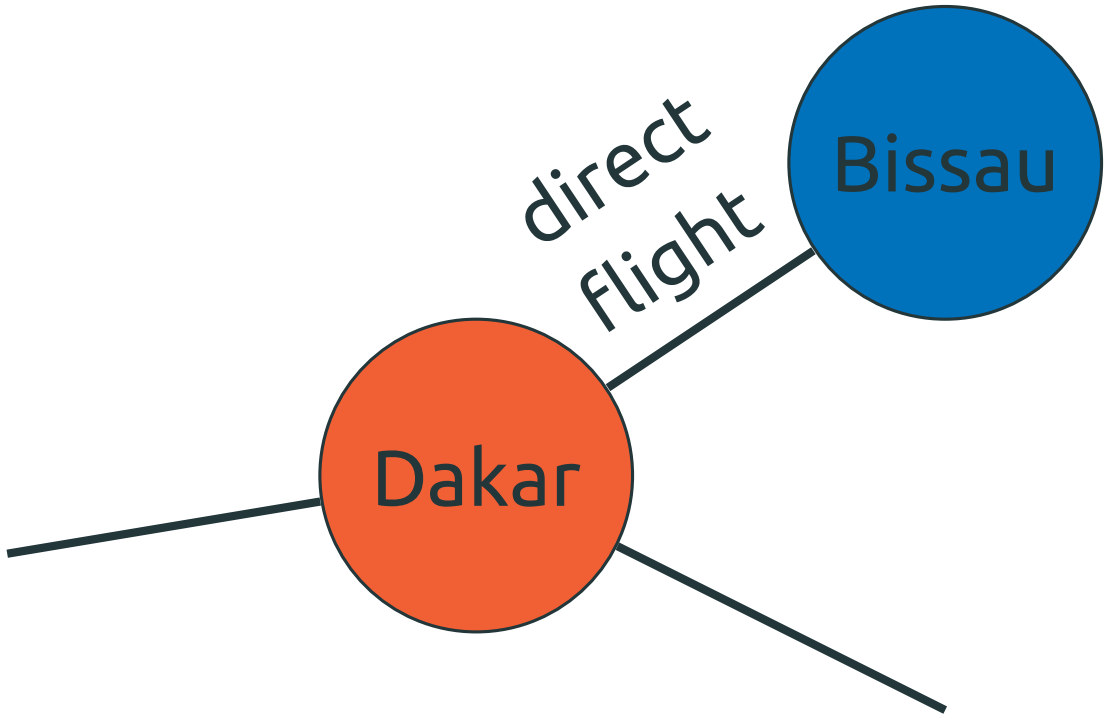
Graph Examples

Graph Applications

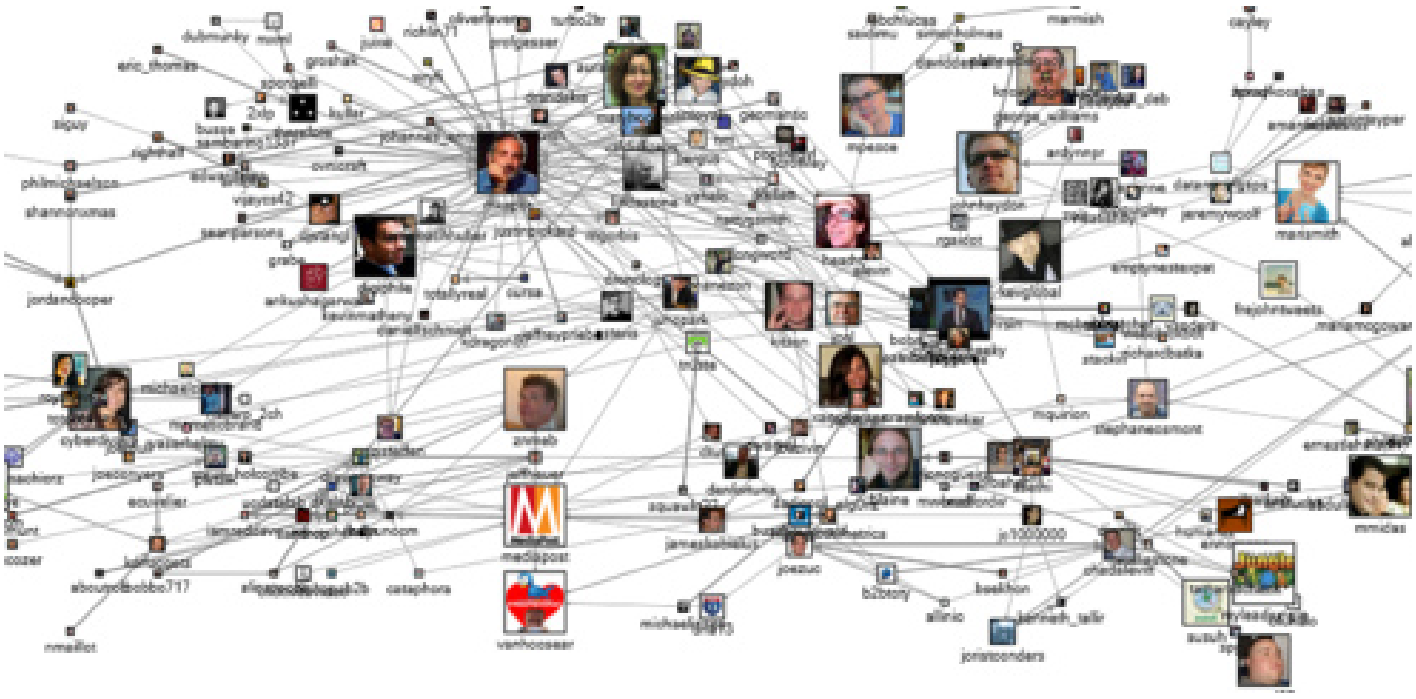
Airlines Graph



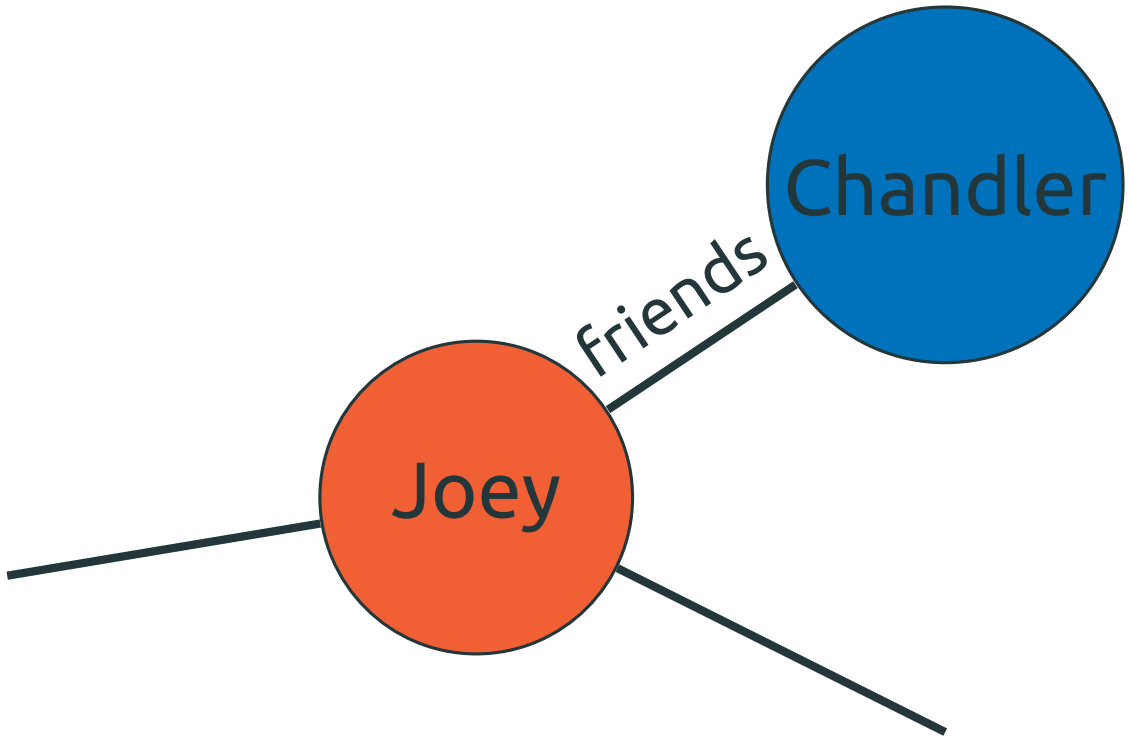
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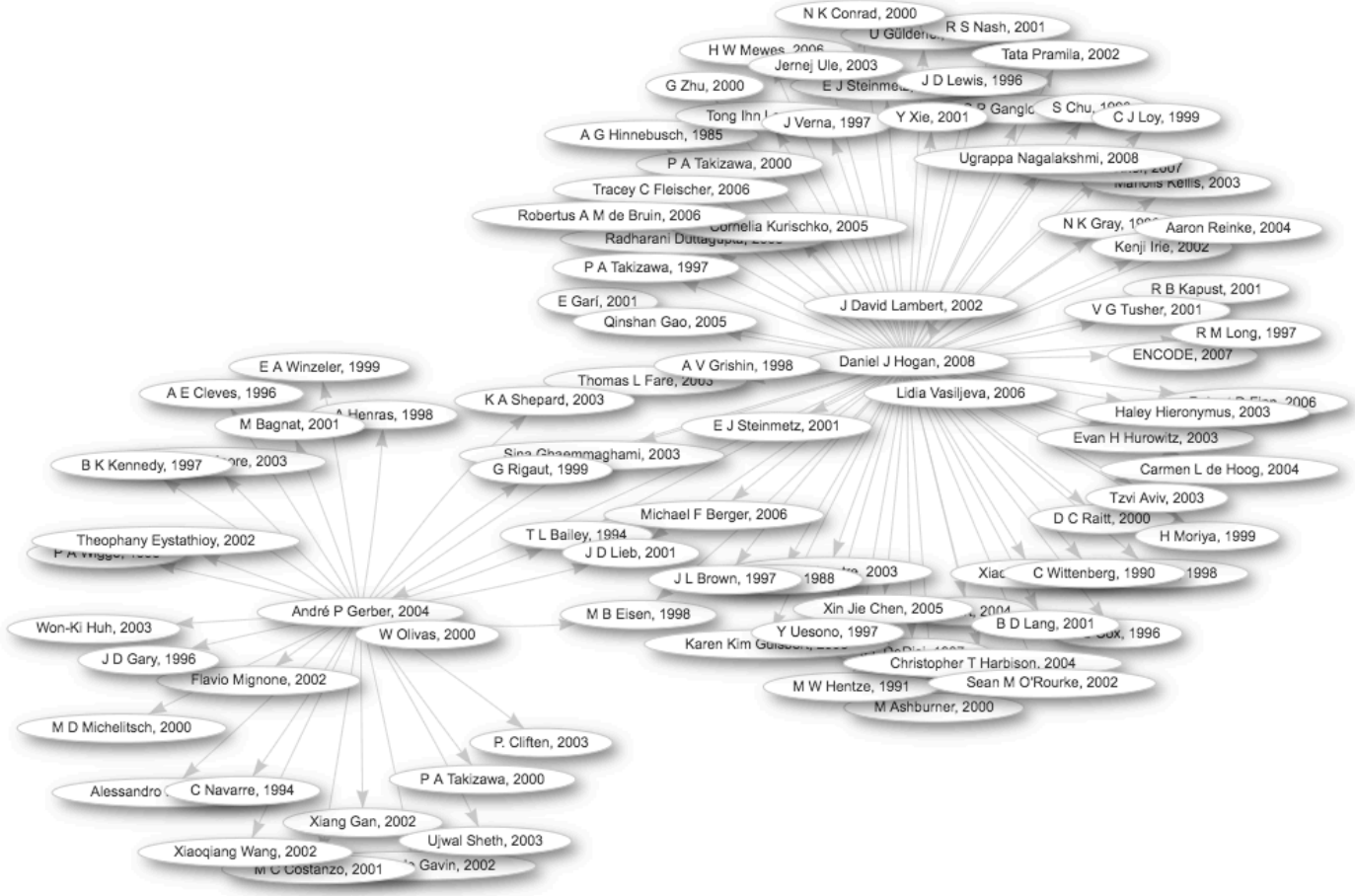
Facebook Graph



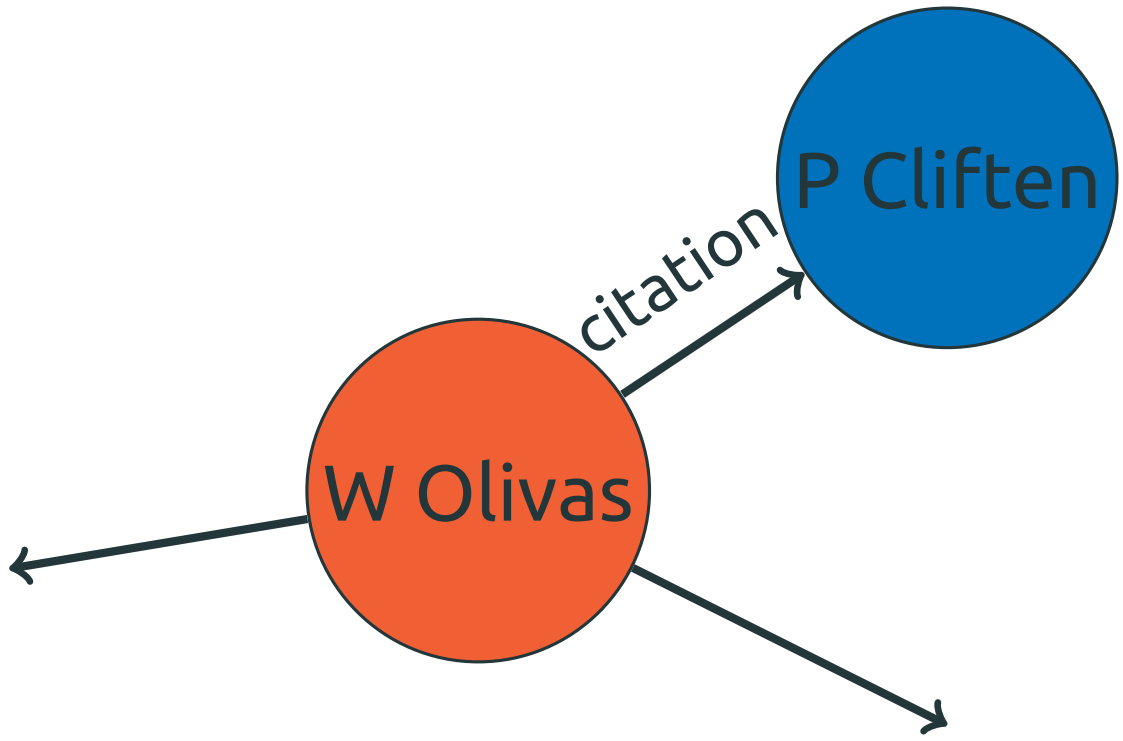
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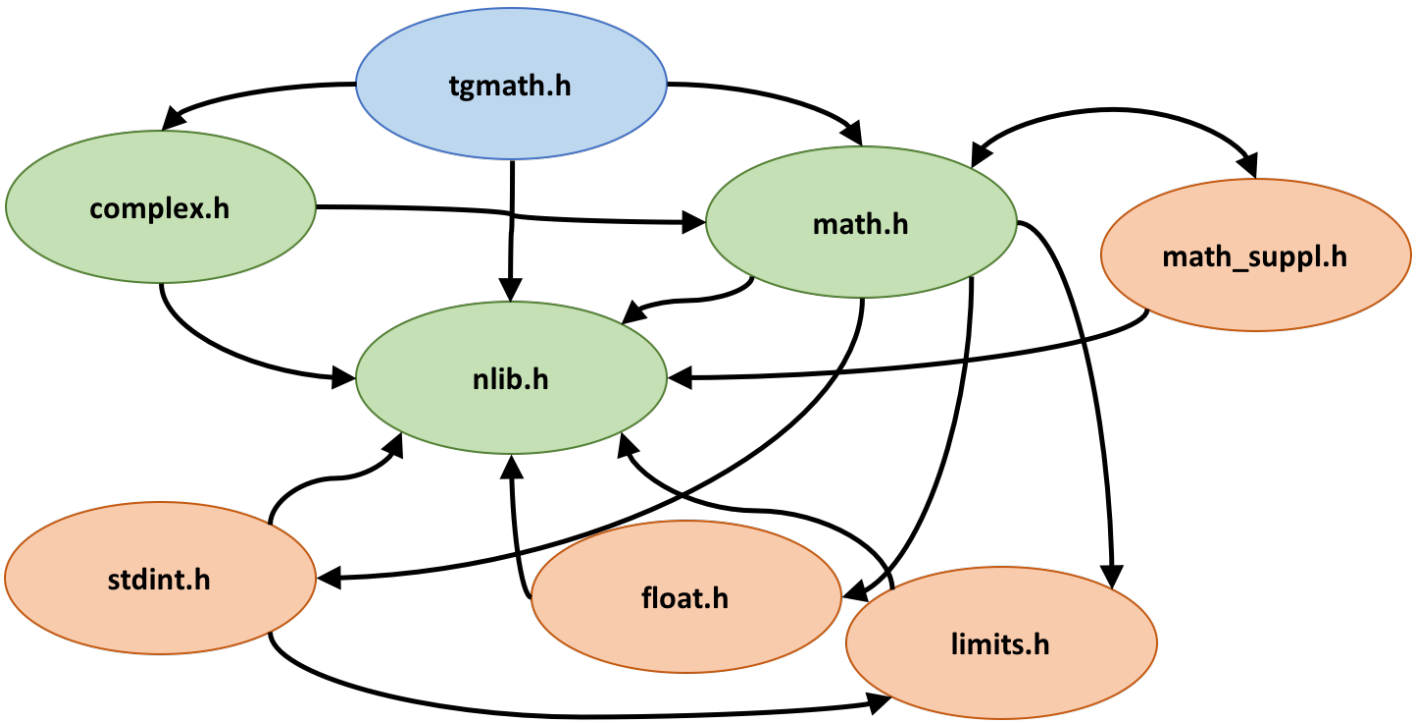
Citation Graph For a Paper



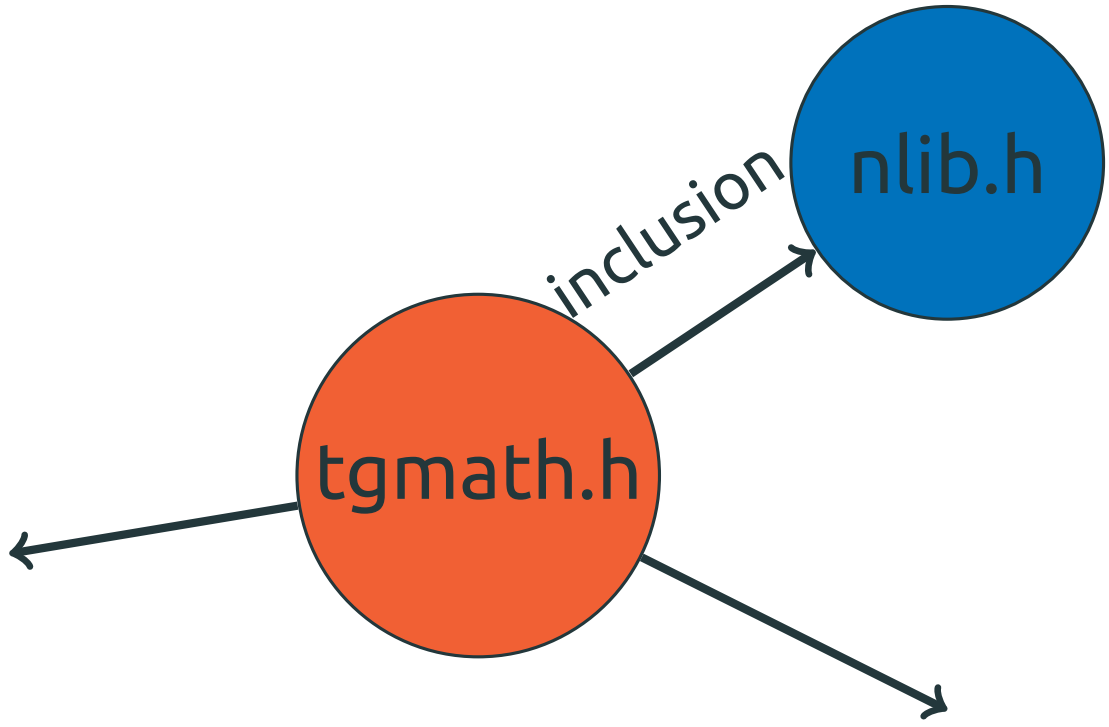
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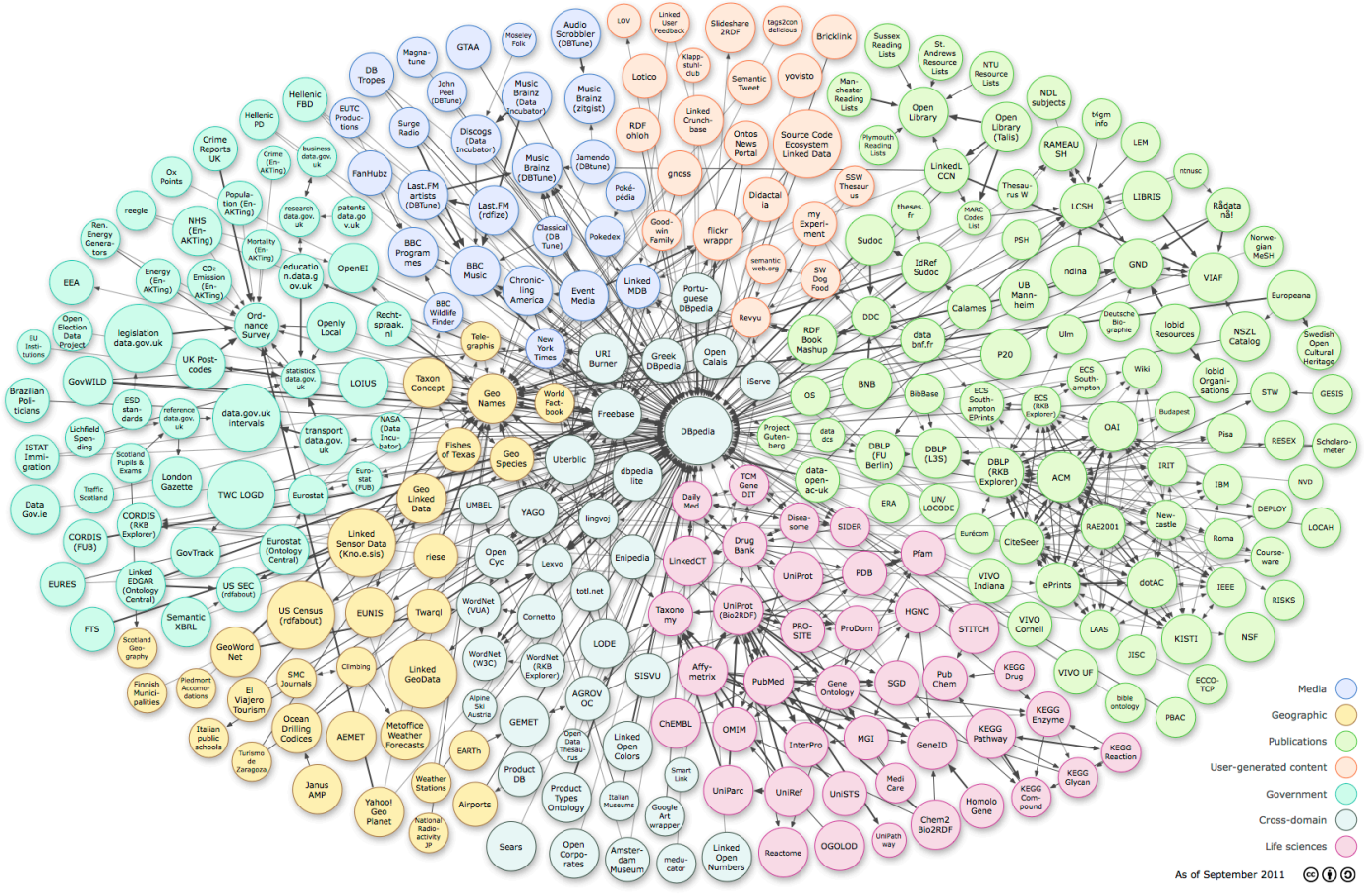
Dependency Graph



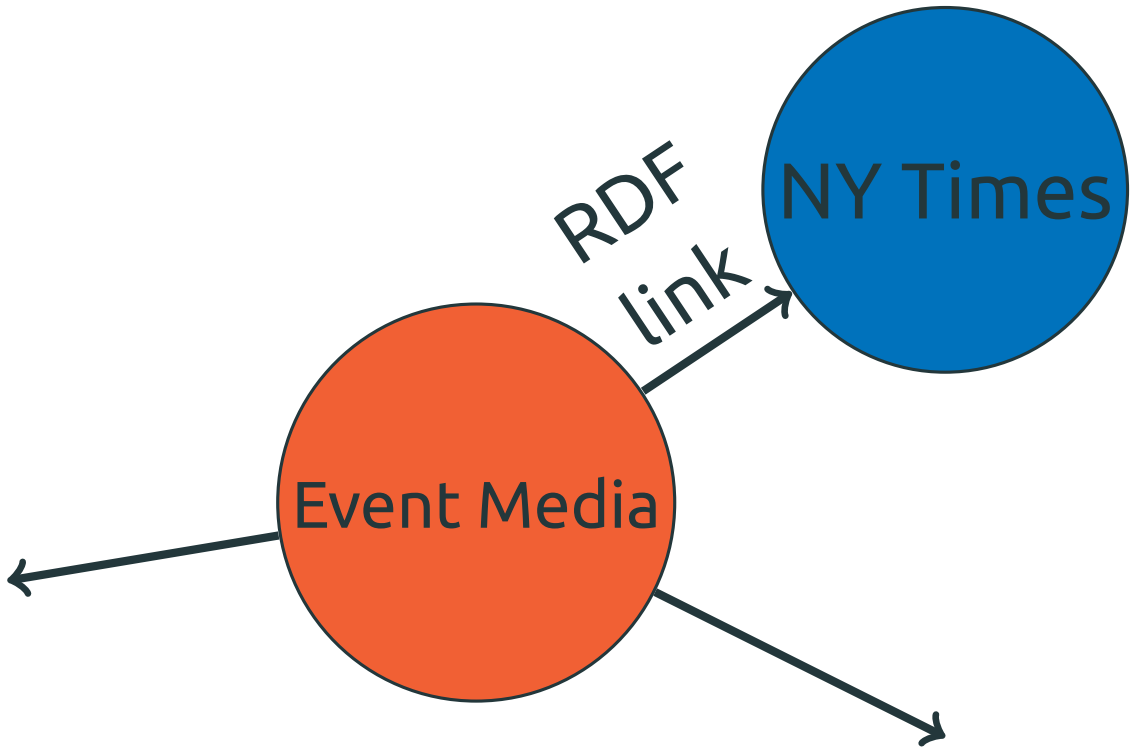
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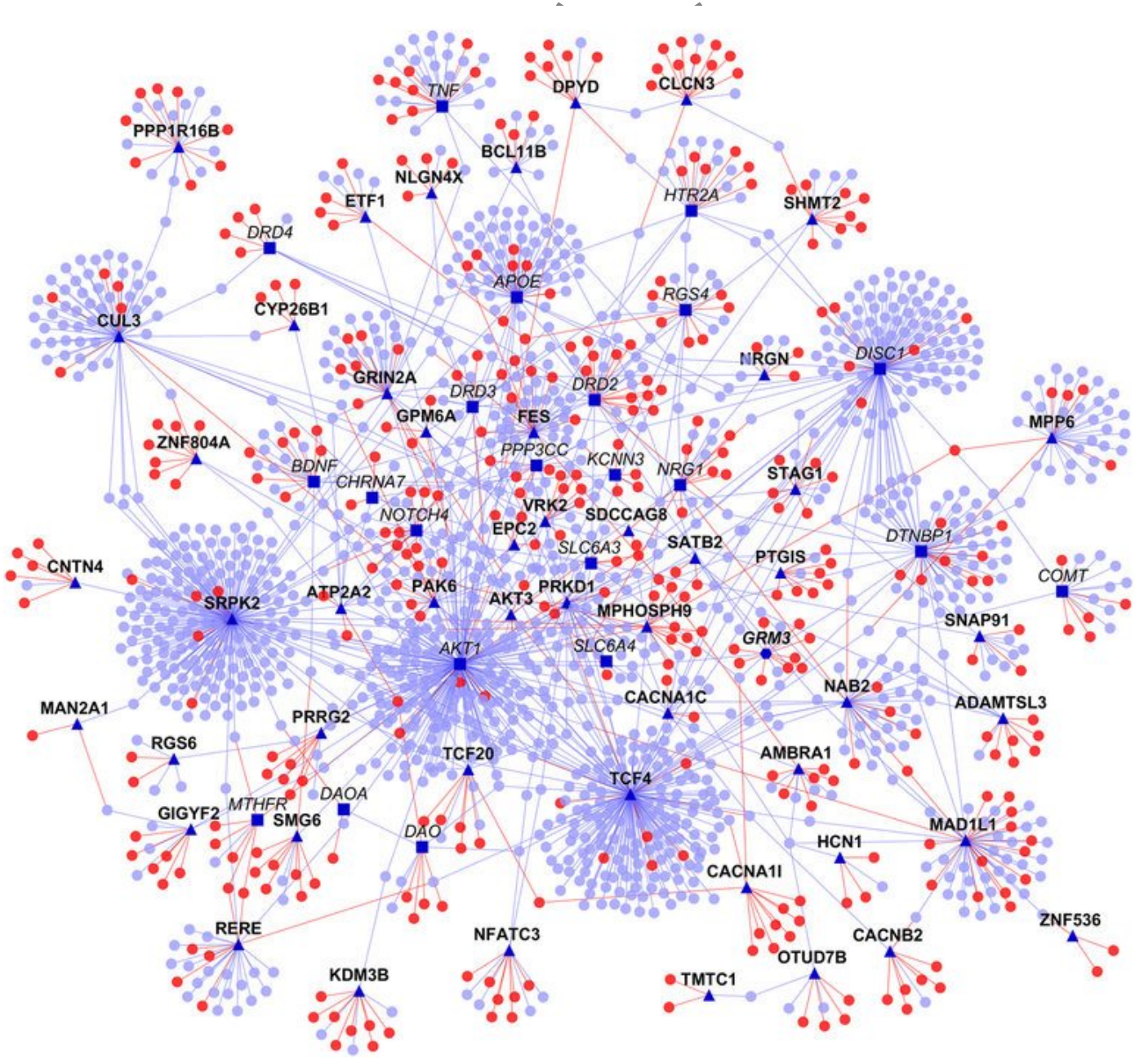
Linked Open Data Diagram



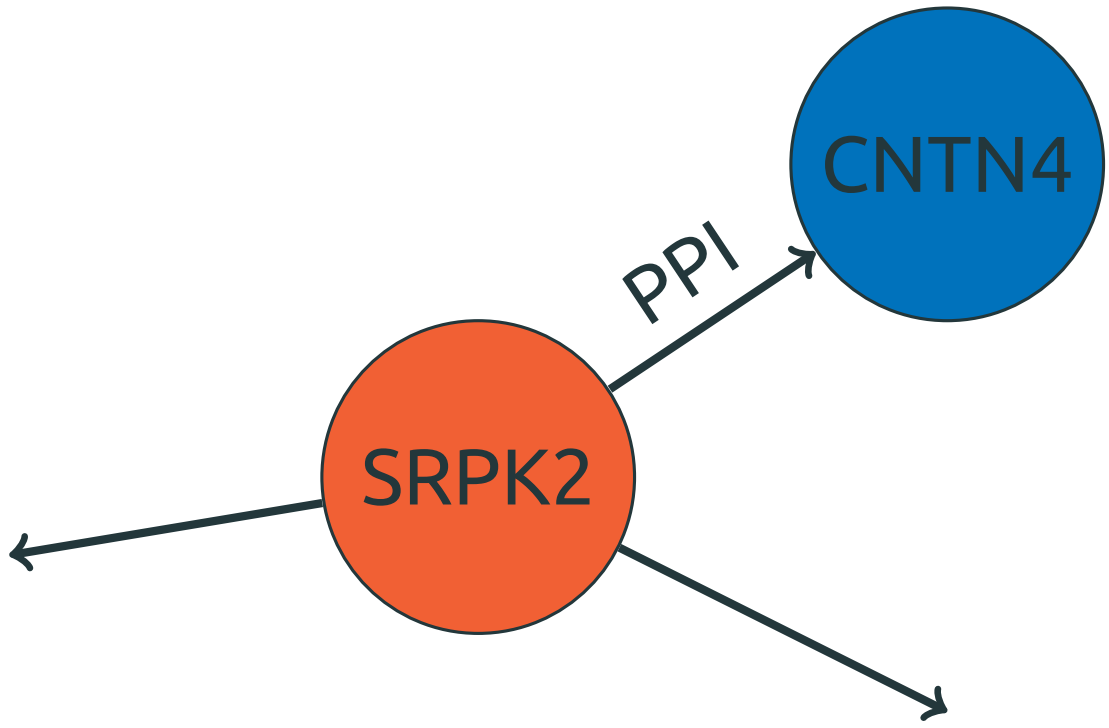
Linked Open Data Diagram



Schizophrenia Protein–Protein Interaction



Schizophrenia Protein–Protein Interaction (PPI)



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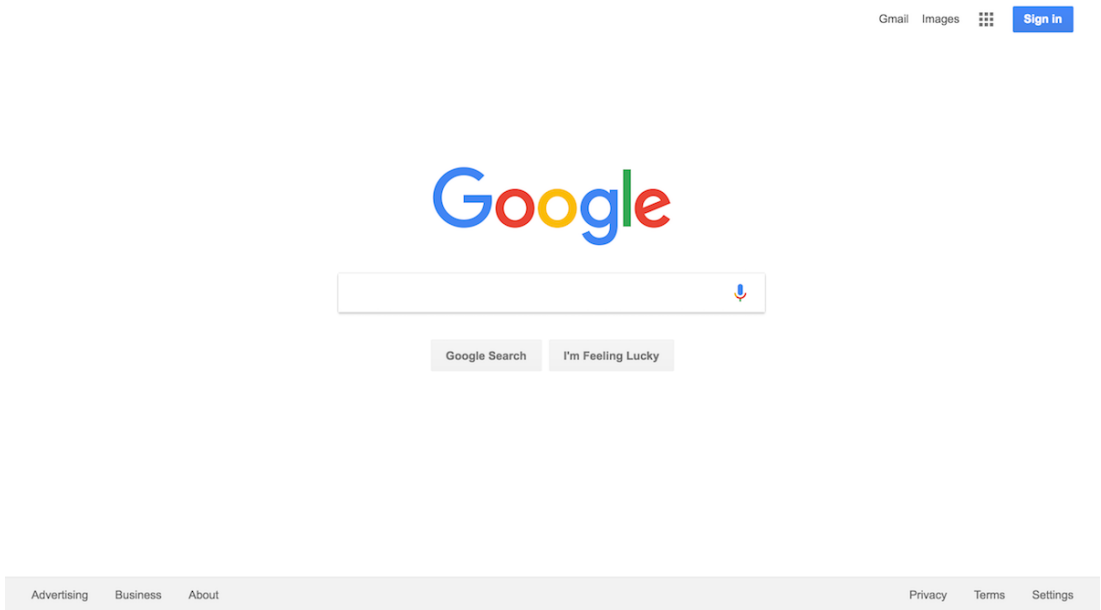
Navigation



Navigation


A
Central Park W
B
Penn Plz, 4
Central Park
E 79th
E 66th St
E 59th St
E 57th St
1st Ave
FDR Dr
Grand Central Terminal
Pennsylvania
Station-New York
Broadway

PageRank


Gmail
Images
Sign in
Google
Google Search
I'm Feeling Lucky
Advertising
Business
About
Privacy
Terms
Settings

PageRank


graph
All
Images
News
Books
Videos
More
Settings
Tools
About 333,000,000 results (1.94 seconds)
graph[1]
/graf/
noun
1. a diagram showing the relation between variable quantities, typically of two variables, each measured along one of a pair of axes at right angles.
synonyms: chart, diagram; More
verb
1. plot or trace on a graph.
synonyms: plot, trace, draw up, delineate
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Translations, word origin, and more definitions
Feedback
Graph - Wikipedia
https://en.wikipedia.org/wiki/Graph
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Graph of a function - Wikipedia
https://en.wikipedia.org/wiki/Graph_of_a_function

PageRank


Graph TV
graphtv.kevinformatics.com/
Graph Ratings of Your Favorite TV Shows. Visualize IMDb ratings and trends of TV shows by episode.
Have you seen Mad Men, Breaking Bad, or Battlestar ...
Previous
1 2 3 4 5 6 7 8 9 10
Next

PageRank


graph
All
Images
News
Books
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More
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noun
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Graph - Wikipedia
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Graph of a function - Wikipedia
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Game Strategies

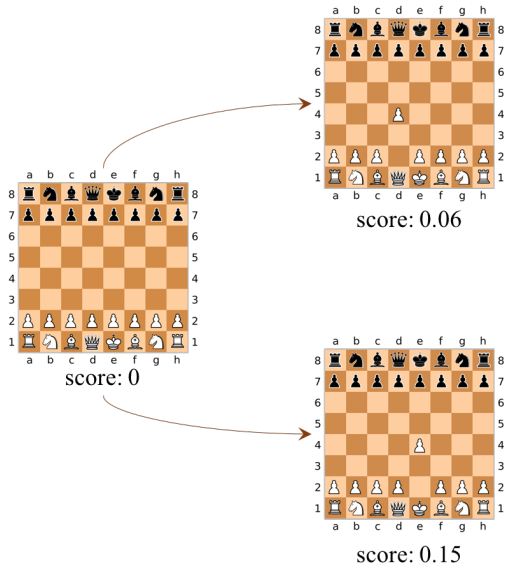


Game Strategies

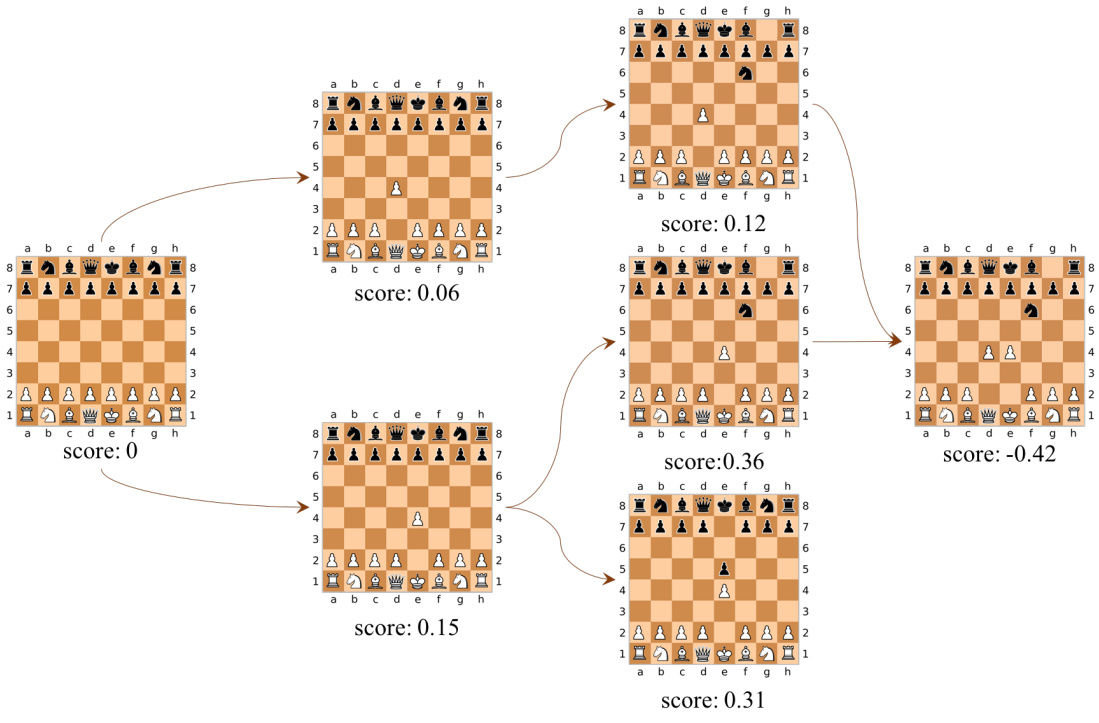


score: 0

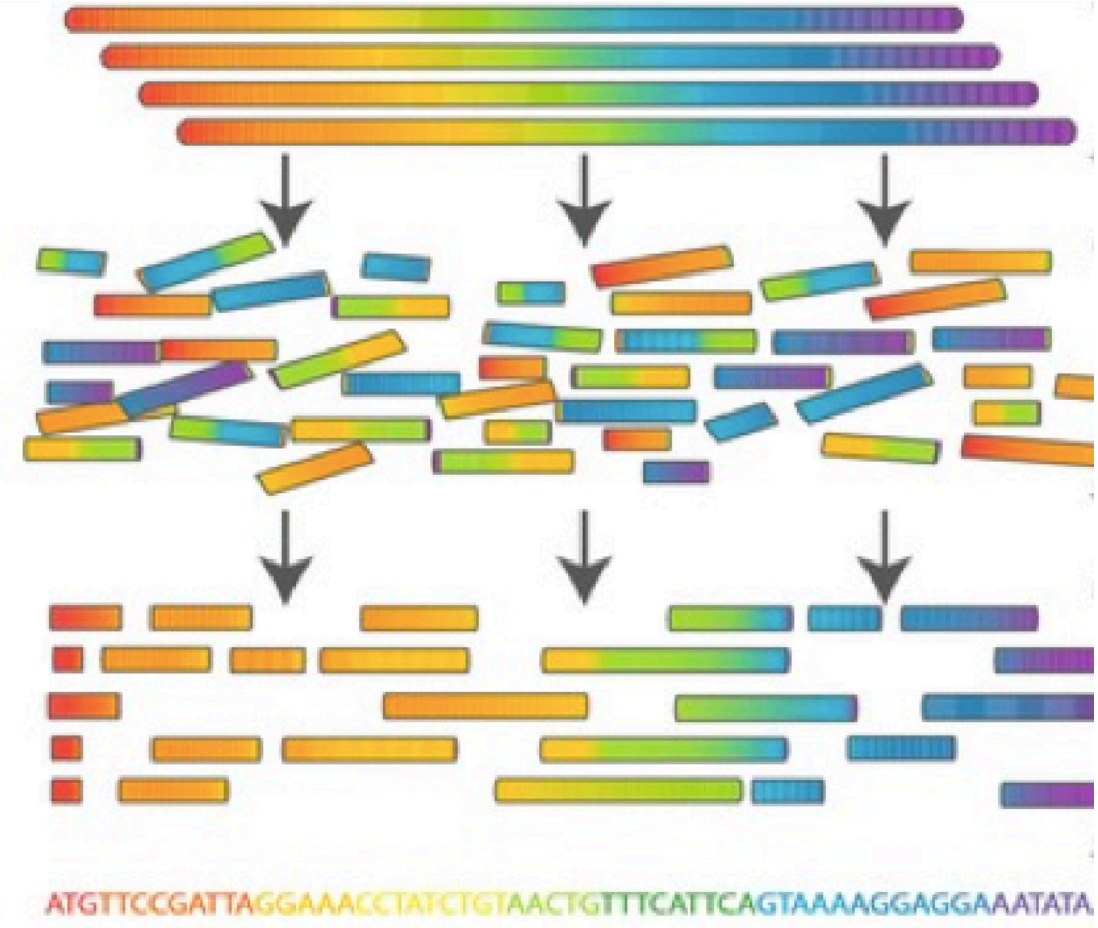
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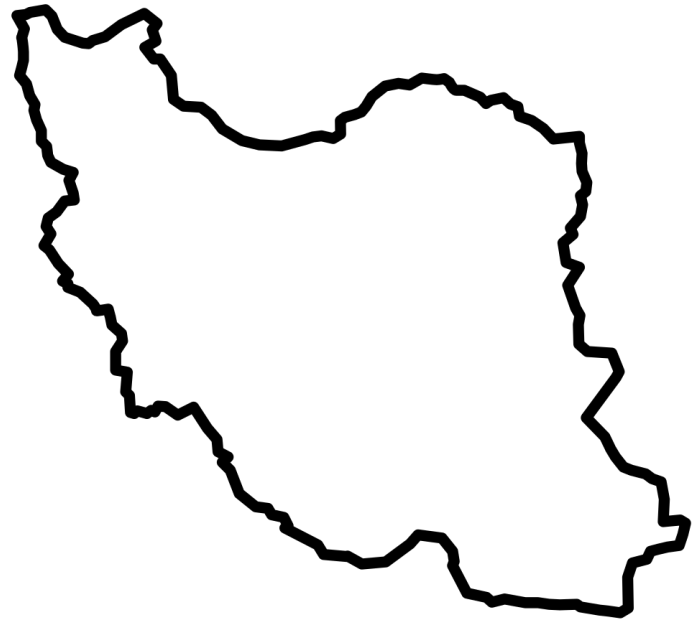
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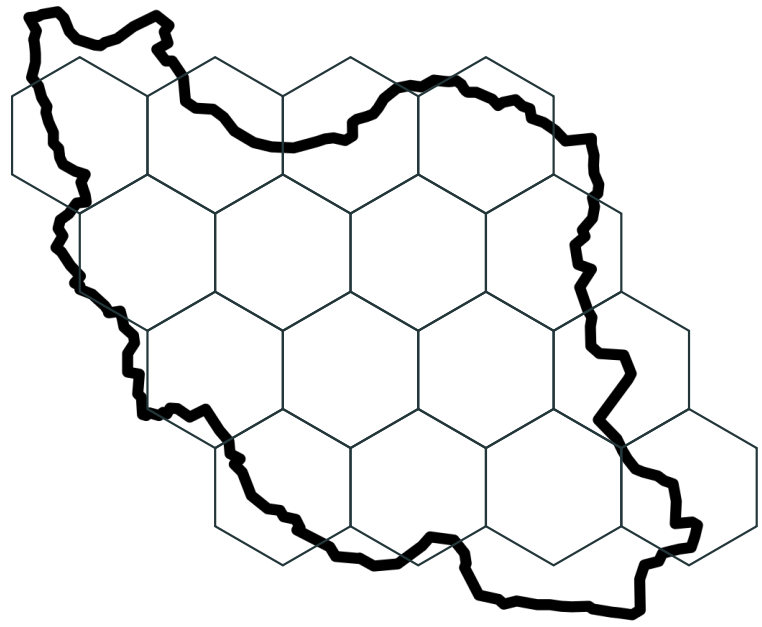
Genome Assembly



GSM

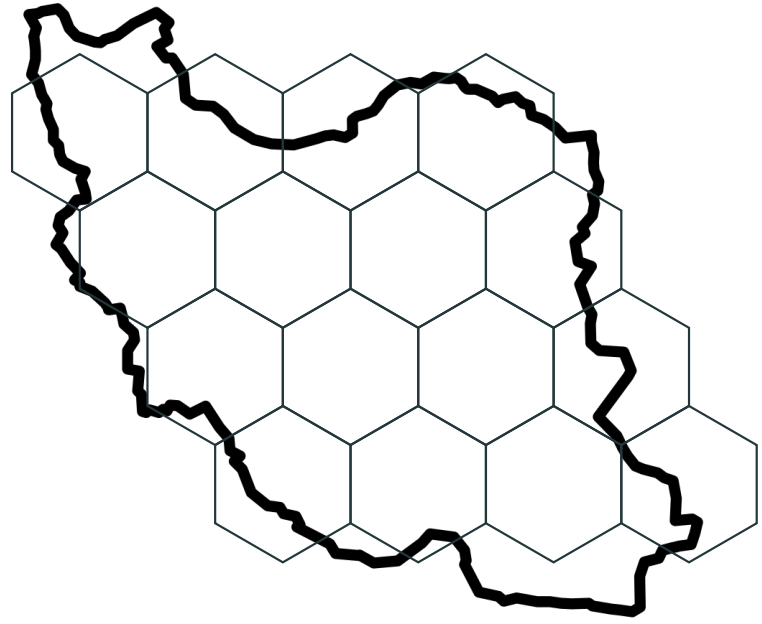


GSM



GSM

4 Frequency Ranges

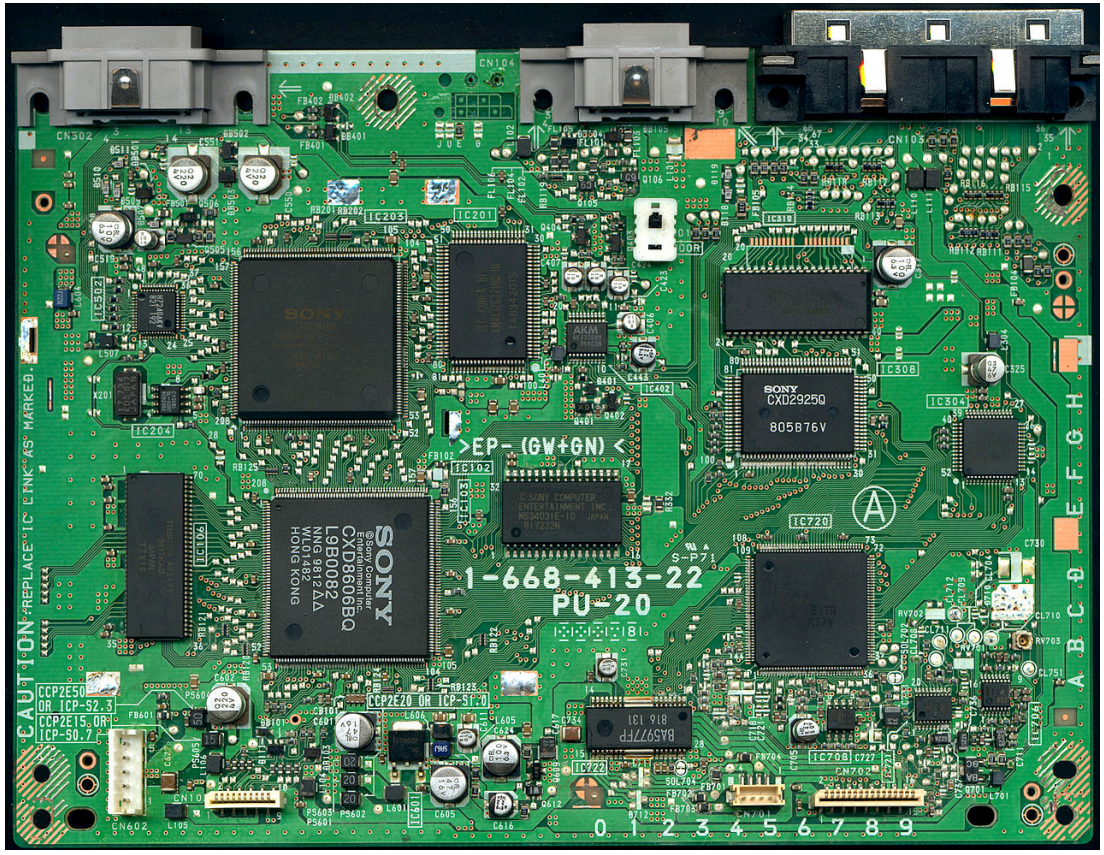


GSM

4 Frequency Ranges



Computer Chips



Computer Chips

